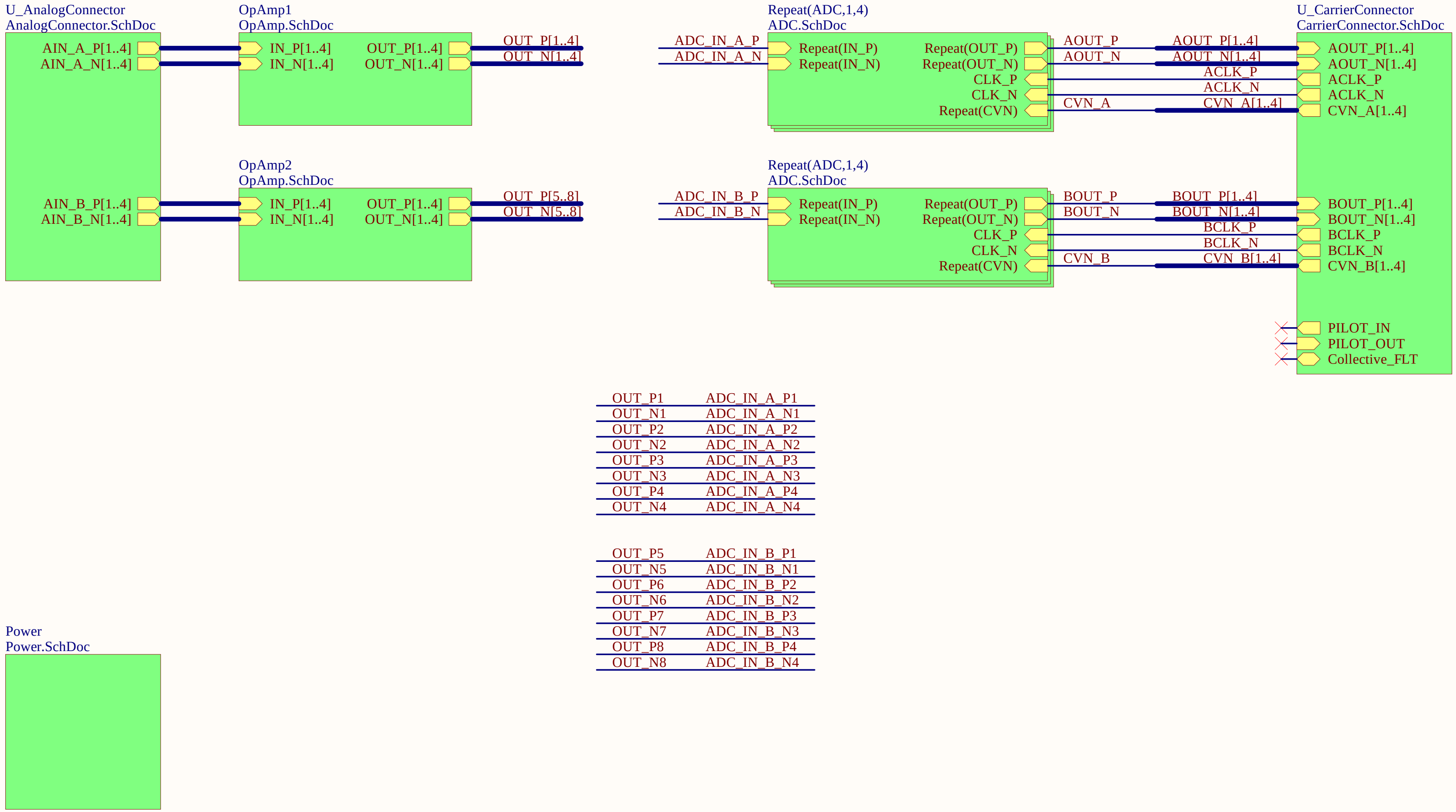


TUM EAL

Title: **UltraZohm ADC** (TopSheet.SchDoc)

Size: A4 Revision: 3v03

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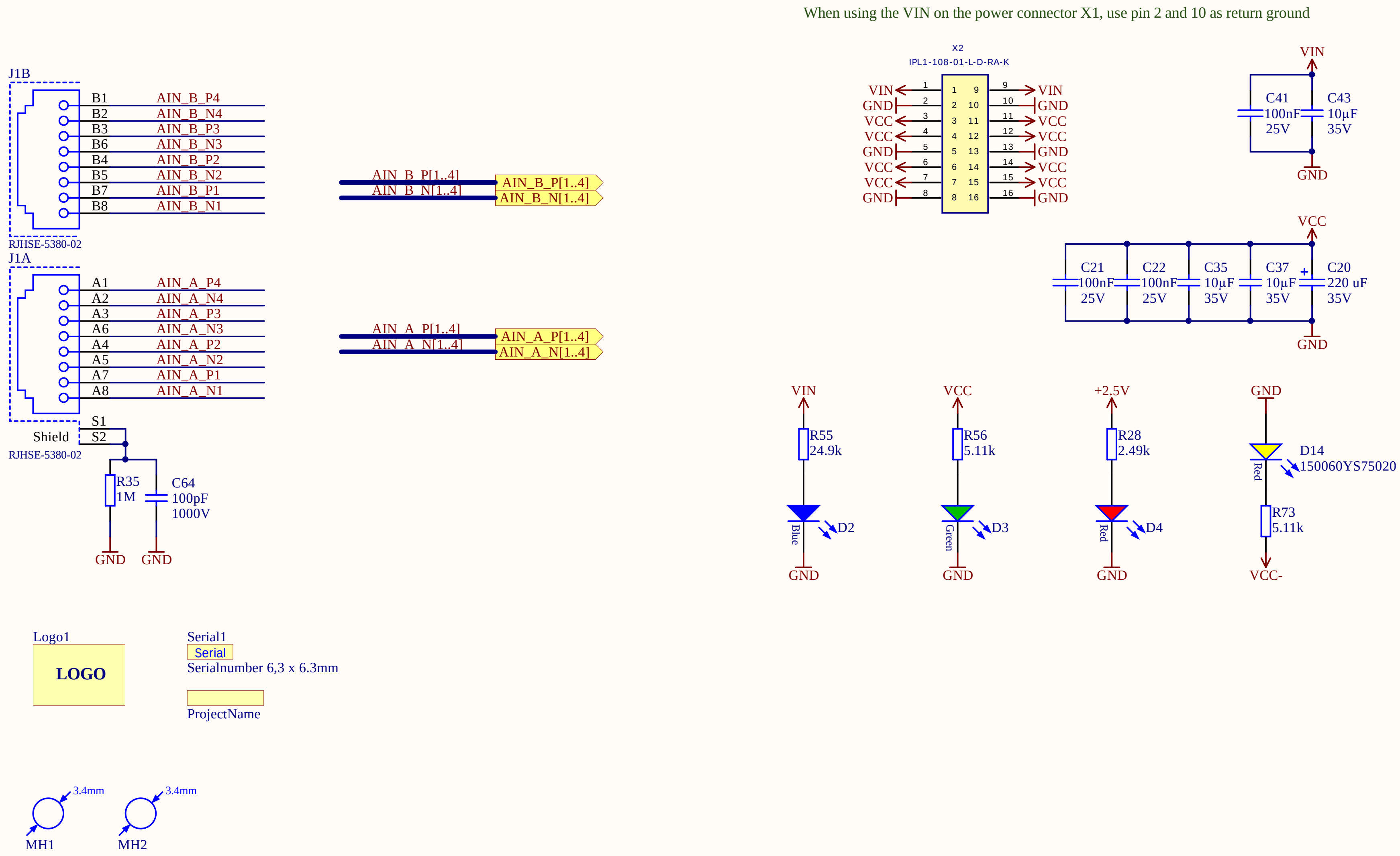
*Arcisstraße 21*
*80333 München*
*GERMANY*

Date: 27.07.2020 Time: 10:18:30 Sheet 1 of 14 Author: Simon Lukas

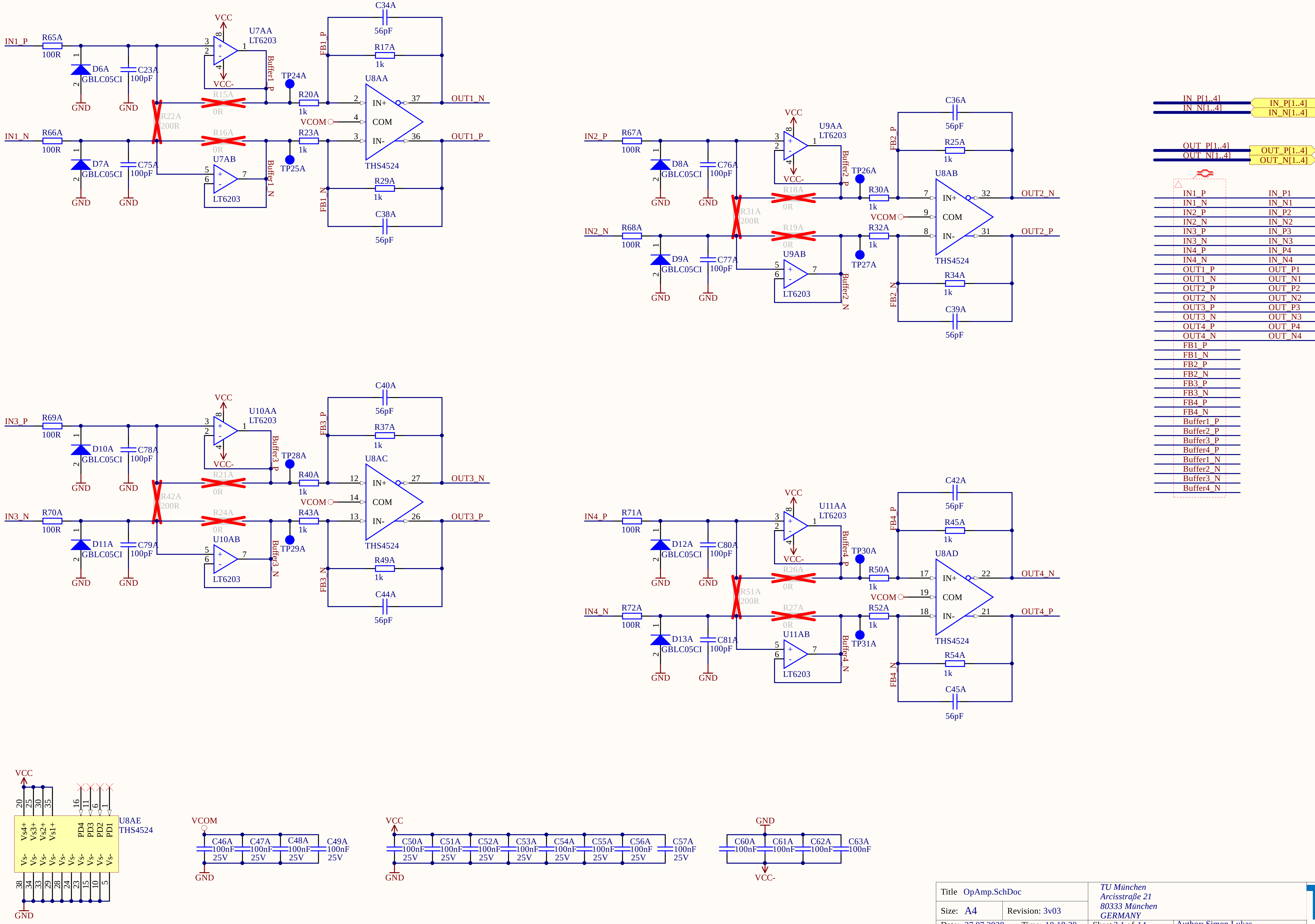
Project: UltraZohm_ADC.PrjPCB




TUM EAL

| | | |
|---|---|---|
| Title AnalogConnector.SchDoc | | TU München<br>Arcisstraße 21<br>80333 München<br>GERMANY |
| Size: A4 | Revision: 3v03 | |
| Date: 27.07.2020 | Time: 10:18:30 | Sheet 2 of 14 — Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | | |

Without buffer: fit R15,R16 with 0R; U7 dnp
With buffer: R15,R16 dnp ; U7 fitted
Current signal: R15,R16 dnp ; U7 fitted; R22 measurement shunt

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| Title | OpAmp.SchDoc |
| Size: A4 | Revision: 3v03 |
| Date: 27.07.2020 | Time: 10:18:30 |
| Sheet 3.1 of 14 | Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | |

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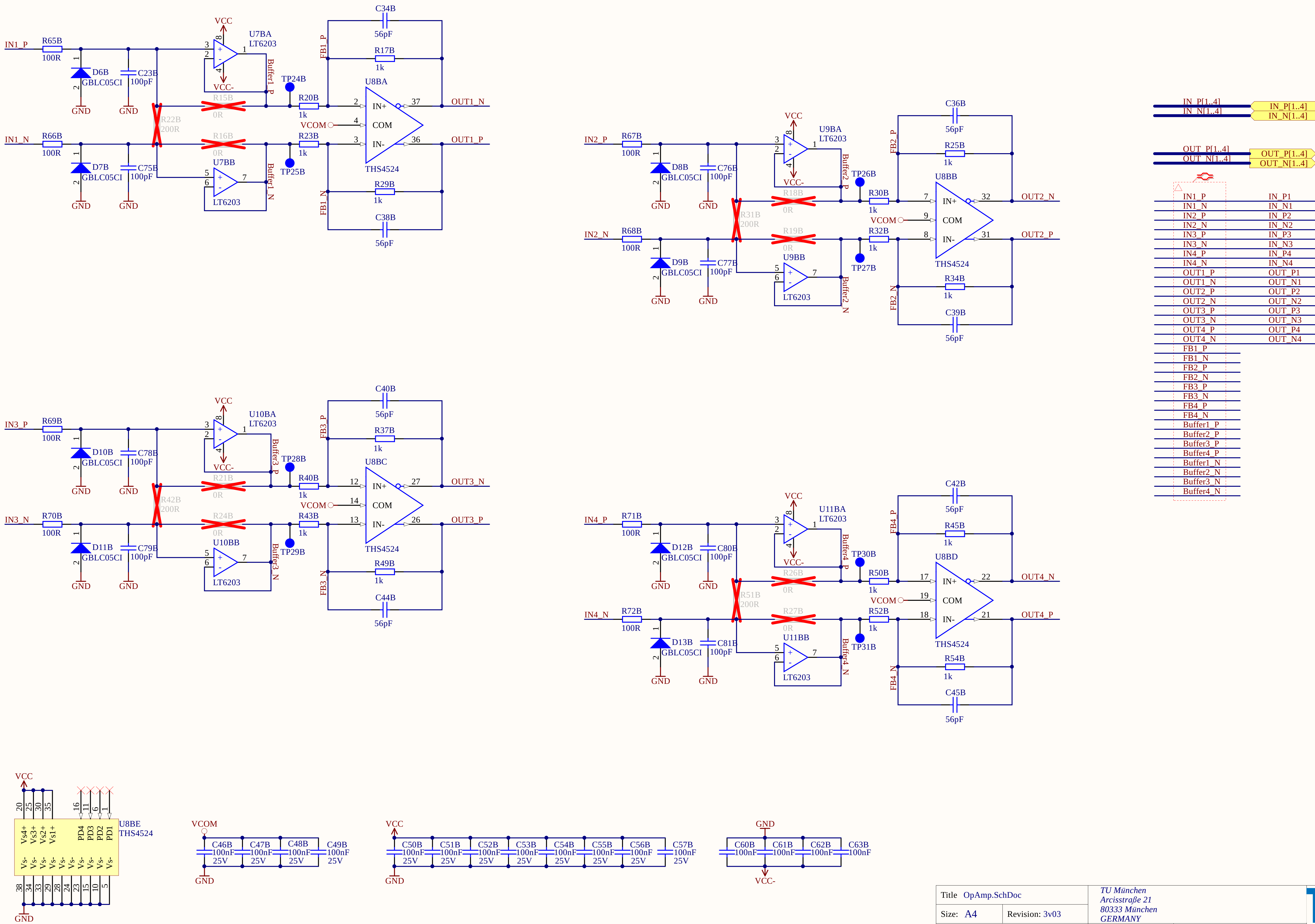
Without buffer: fit R15,R16 with 0R; U7 dnp
With buffer: R15,R16 dnp ; U7 fitted
Current signal: R15,R16 dnp ; U7 fitted; R22 measurement shunt

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| Title | OpAmp.SchDoc |
| Size: A4 | Revision: 3v03 |
| Date: 27.07.2020 | Time: 10:18:31 |
| Sheet 3.2 of 14 | |
| Project: UltraZohm_ADC.PrjPCB | |


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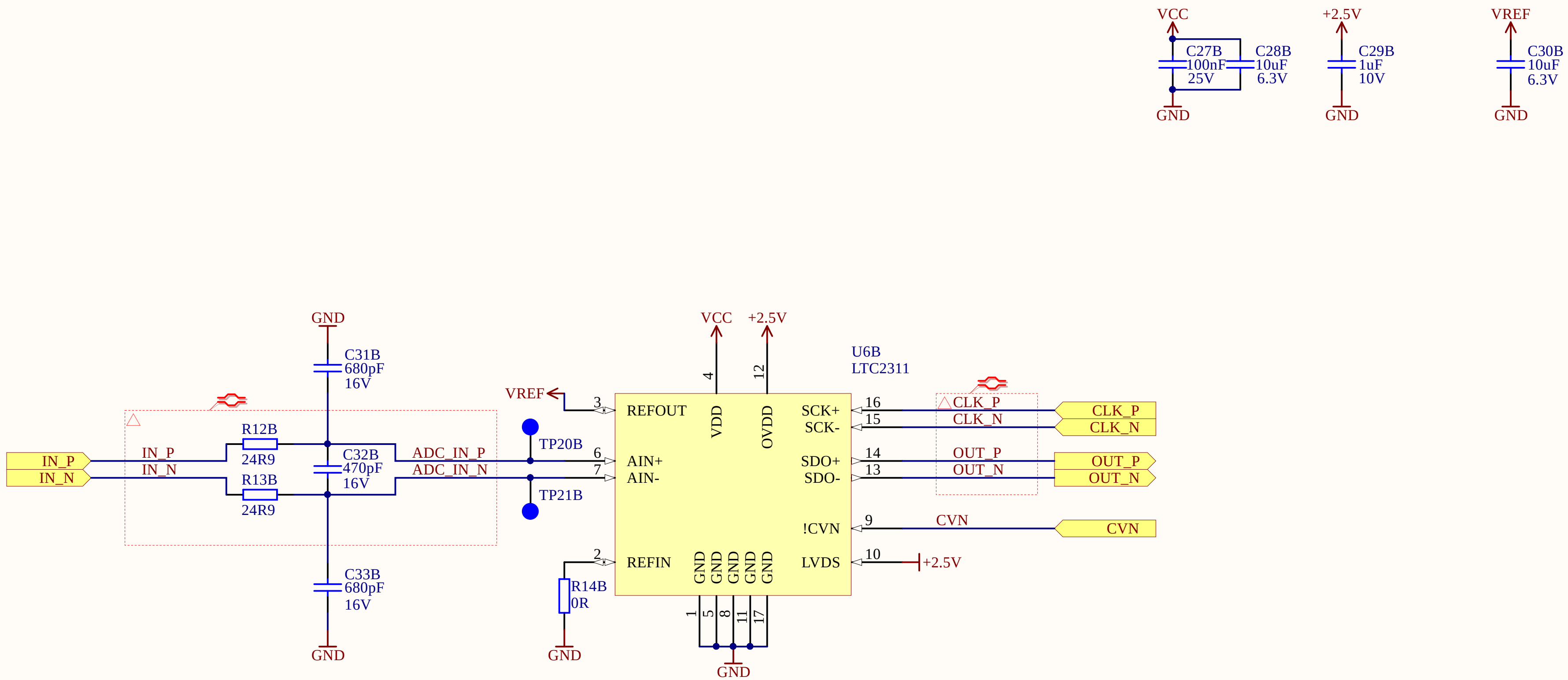
*GERMANY*
Author: Simon Lukas

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor




TUM EAL

Title ADC.SchDoc

Size: A4 Revision: 3v03

Date: 27.07.2020 Time: 10:18:31 Sheet 4.1 of 14

Project: UltraZohm_ADC.PrjPCB

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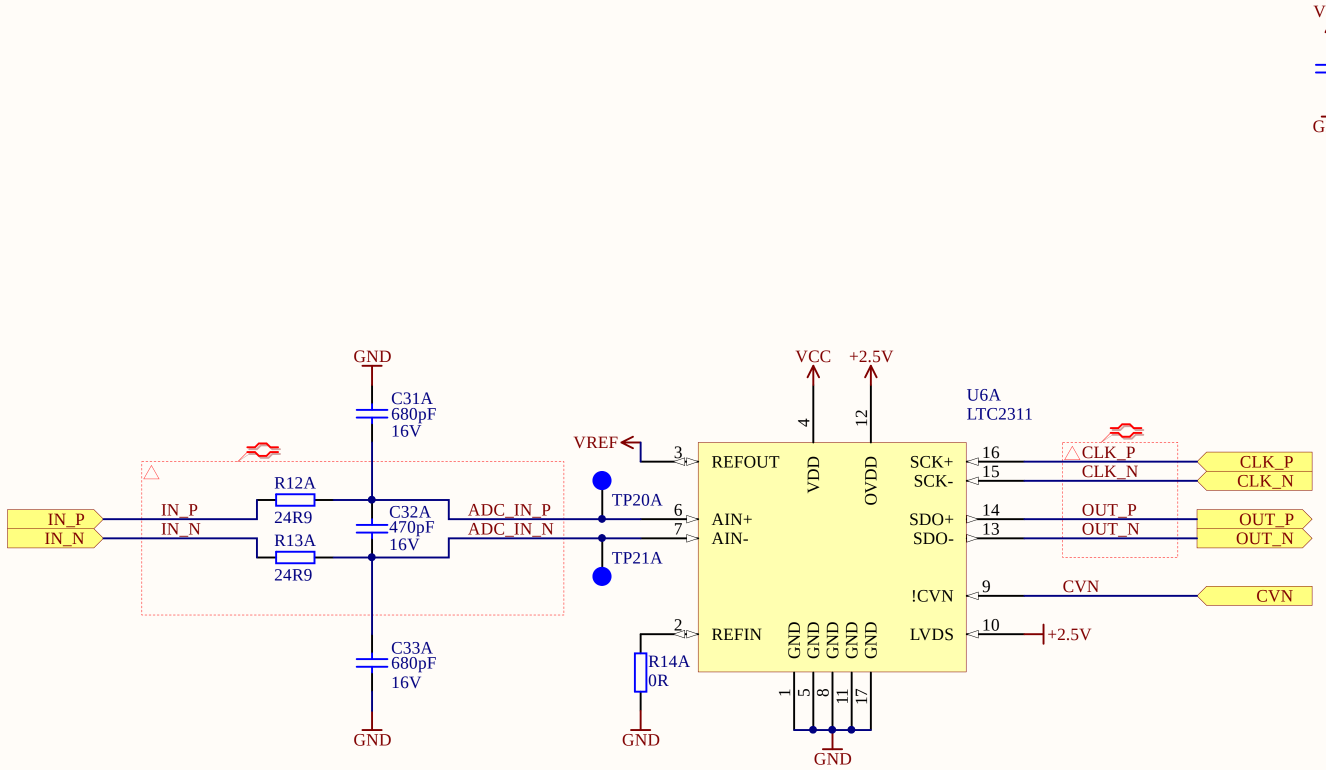
GERMANY
Author: Simon Lukas

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor

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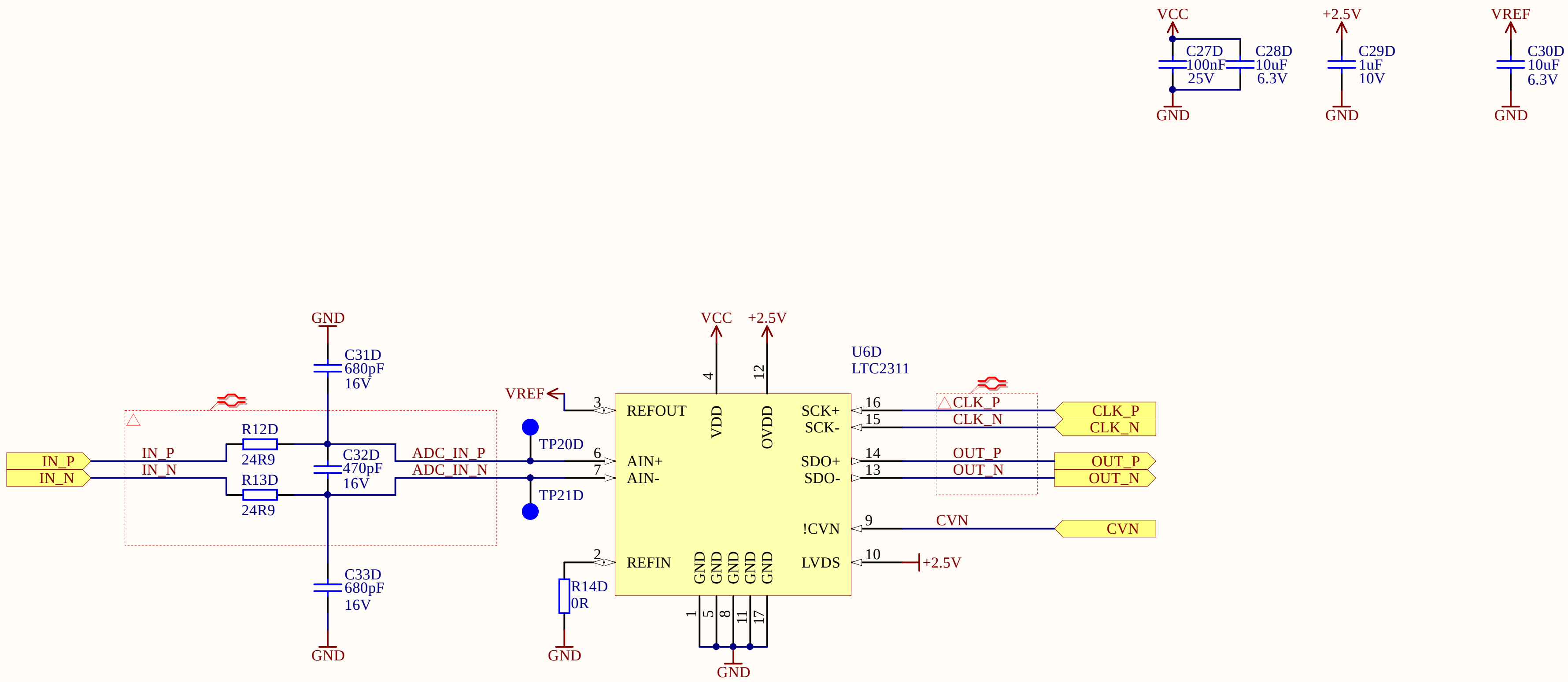
|---|---|---|
| Title ADC.SchDoc | | TUM EAL<br>TU München<br>Arcisstraße 21<br>80333 München<br>GERMANY |
| Size: A4 | Revision: 3v03 | |
| Date: 27.07.2020 | Time: 10:18:31 | Sheet 4.1 of 14 / Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | | |

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor

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| Title ADC.SchDoc | | TUM EAL<br>TU München<br>Arcisstraße 21<br>80333 München<br>GERMANY |
| Size: A4 | Revision: 3v03 | |
| Date: 27.07.2020 | Time: 10:18:31 | Sheet 4.2 of 14 — Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | | |

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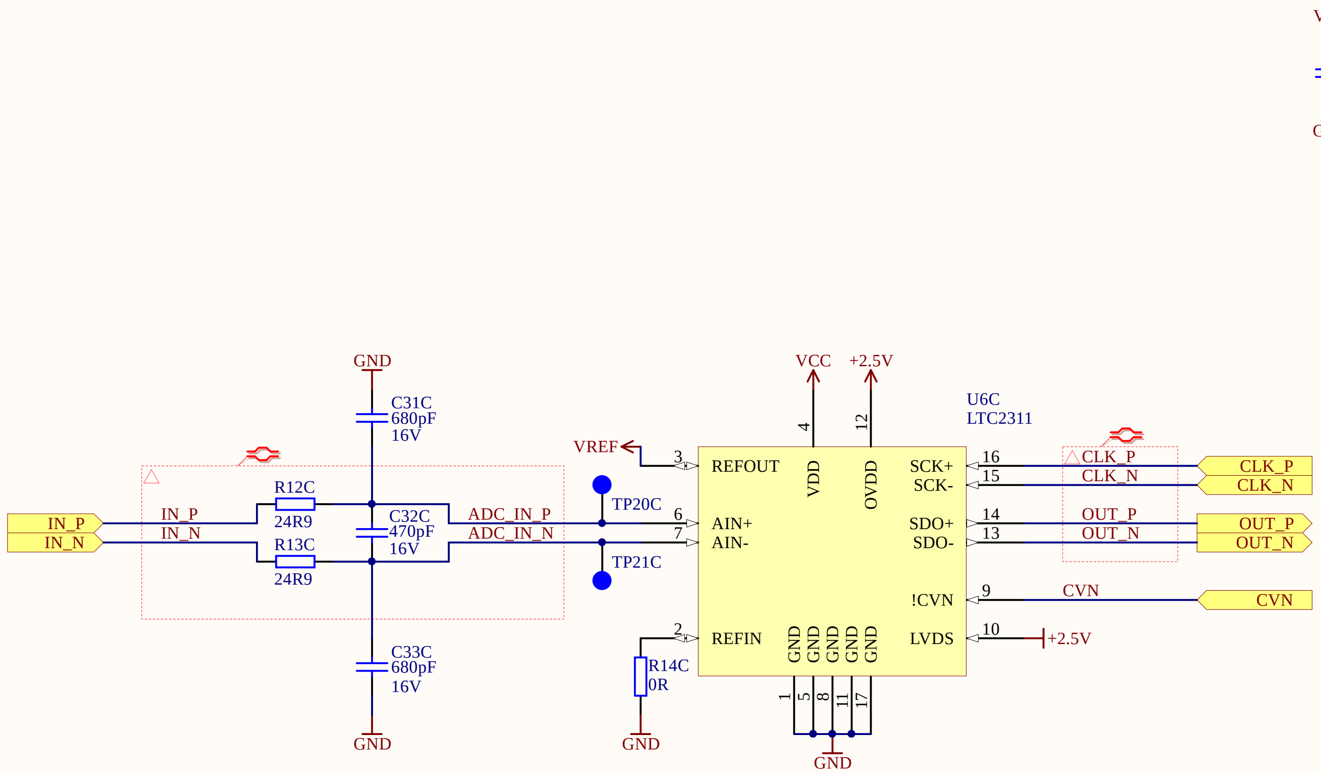


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| Title | ADC.SchDoc |
| Size: A4 | Revision: 3v03 |
| Date: 27.07.2020 | Time: 10:18:31 |
| Sheet 4.2 of 14 | Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | |


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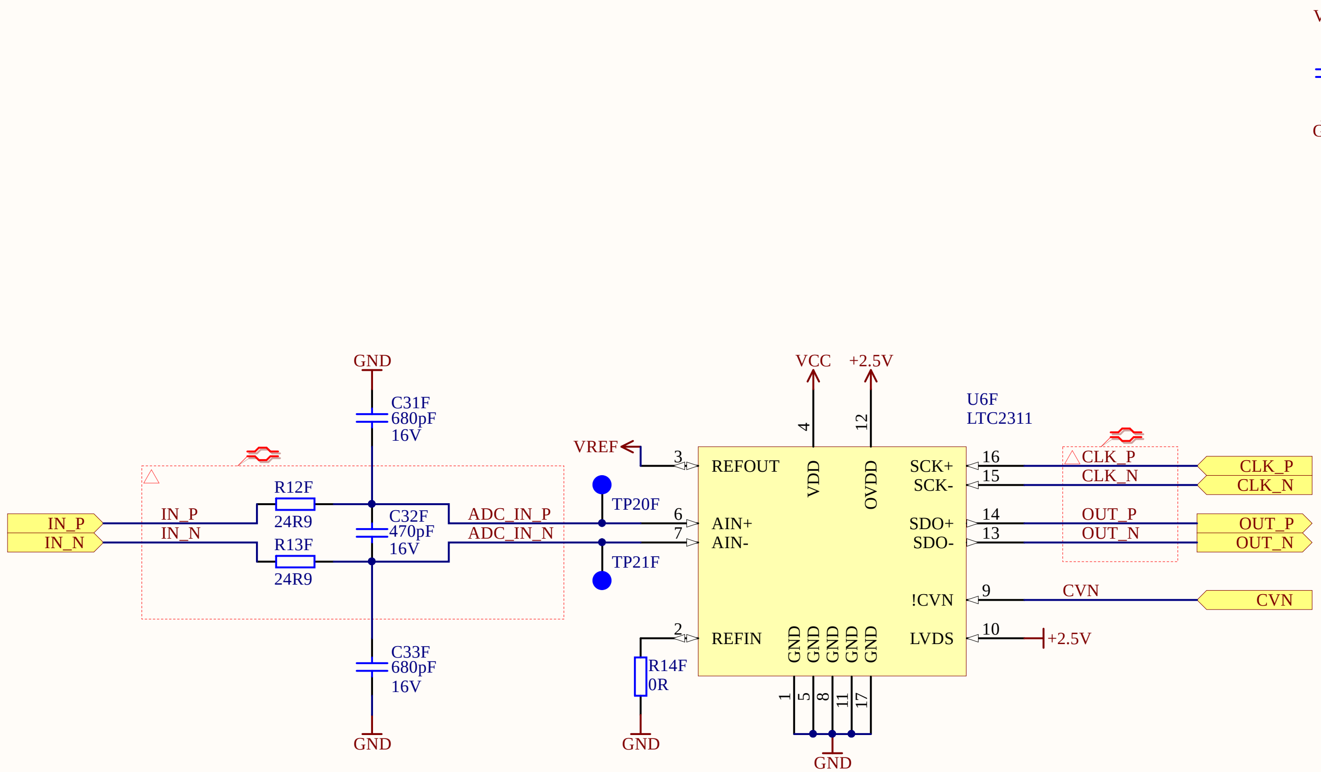
*80333 München*
*GERMANY*

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor

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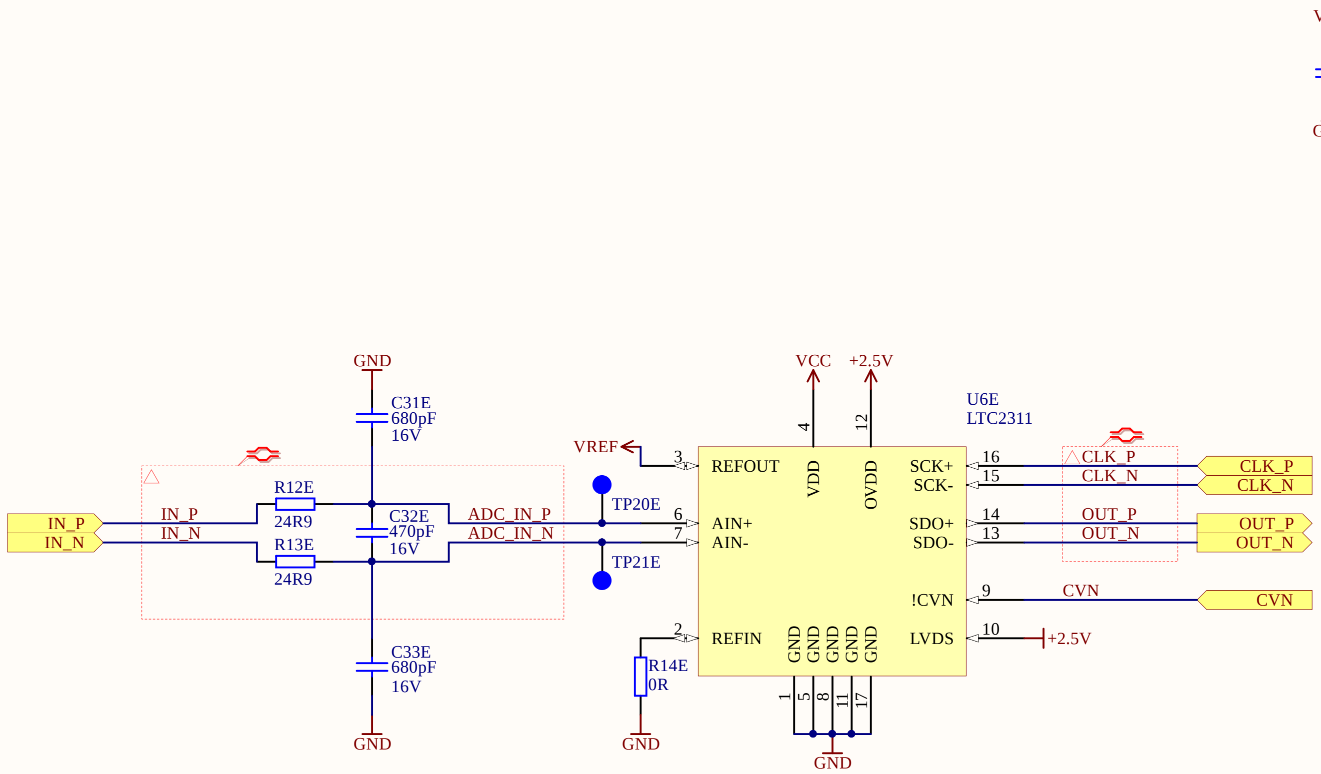
|---|---|---|---|
| Title ADC.SchDoc | | TU München<br>Arcisstraße 21<br>80333 München<br>GERMANY | |
| Size: A4 | Revision: 3v03 | | |
| Date: 27.07.2020 | Time: 10:18:31 | Sheet 4.3 of 14 | Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | | | |

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor




Title ADC.SchDoc

Size: A4 Revision: 3v03

Date: 27.07.2020 Time: 10:18:32 Sheet 4.3 of 14

Project: UltraZohm_ADC.PrjPCB

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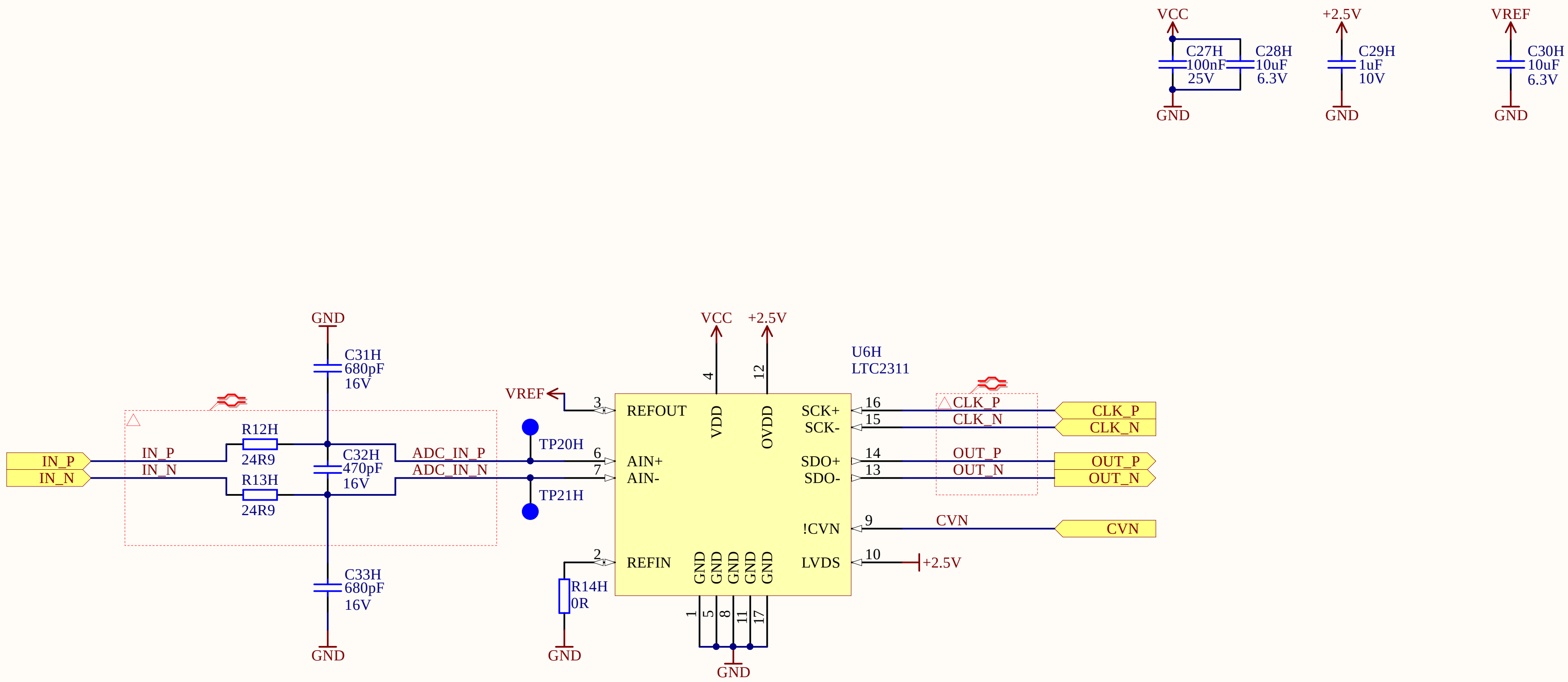
80333 München
GERMANY
Author: Simon Lukas

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor

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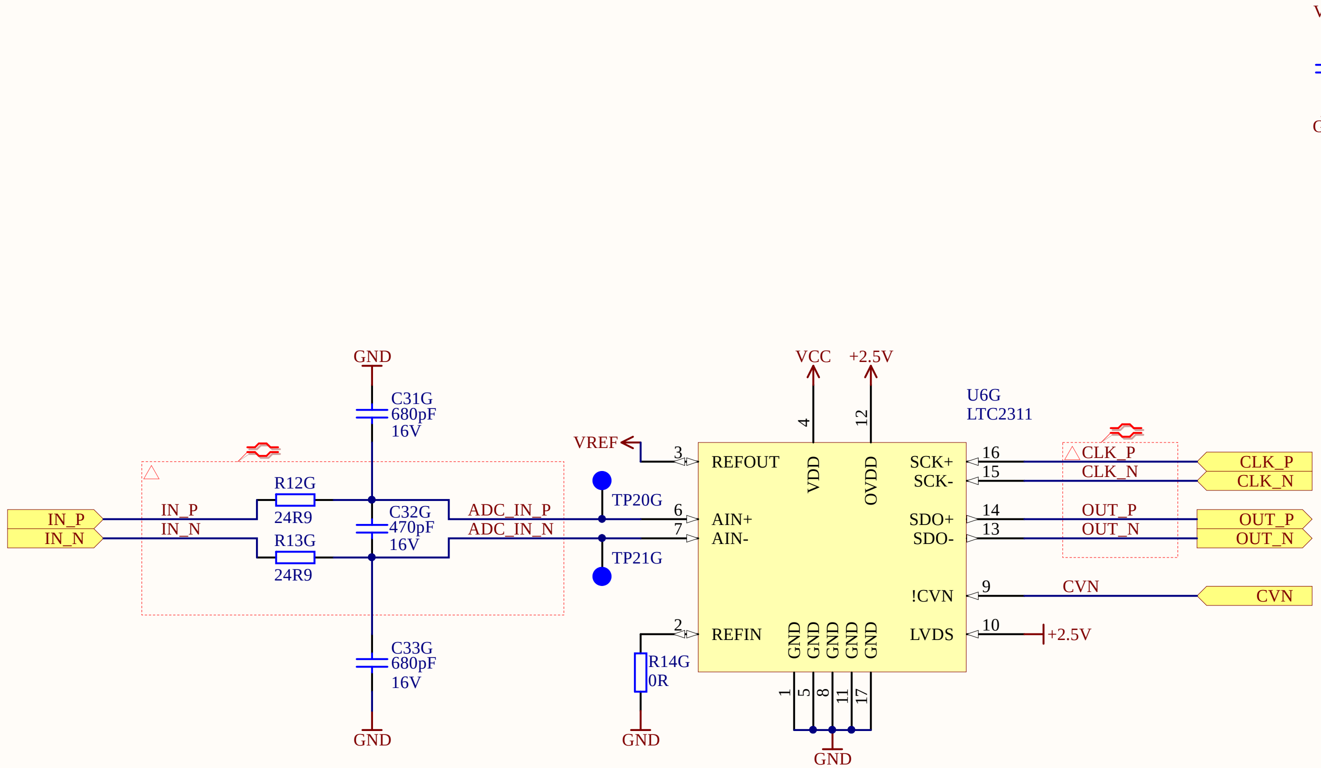
| Size: A4 | Revision: 3v03 | | |
| Date: 27.07.2020 | Time: 10:18:32 | Sheet 4.4 of 14 | Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | | | |

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor

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|---|---|---|---|
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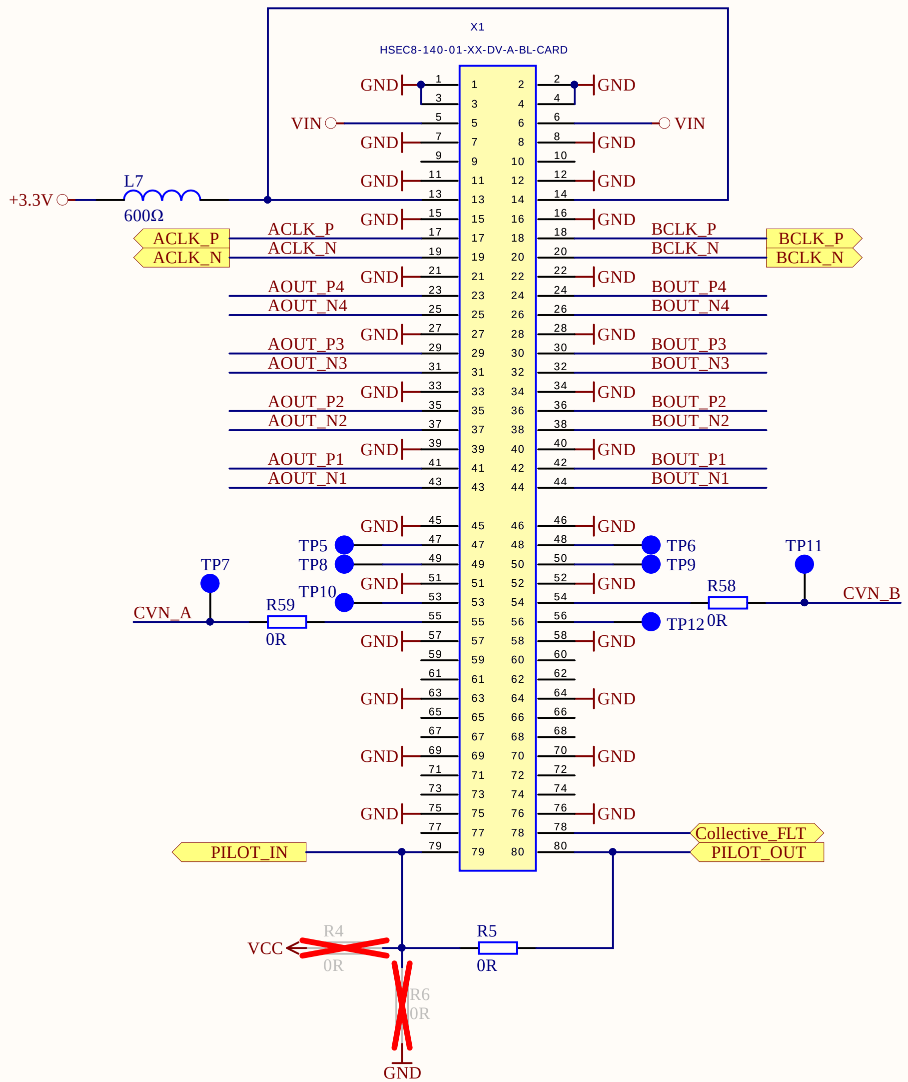
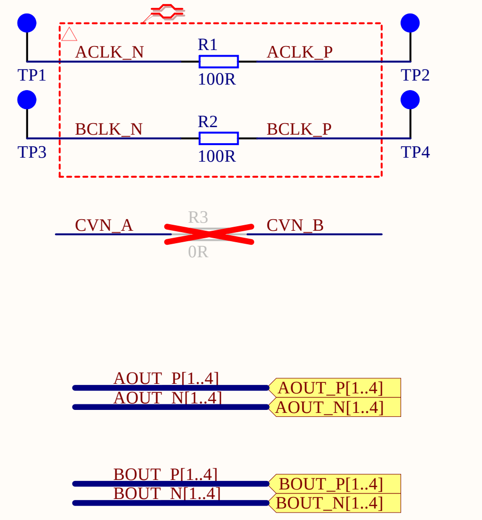
| Size: A4 | Revision: 3v03 | | |
| Date: 27.07.2020 | Time: 10:18:32 | Sheet 4.4 of 14 | Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | | | |

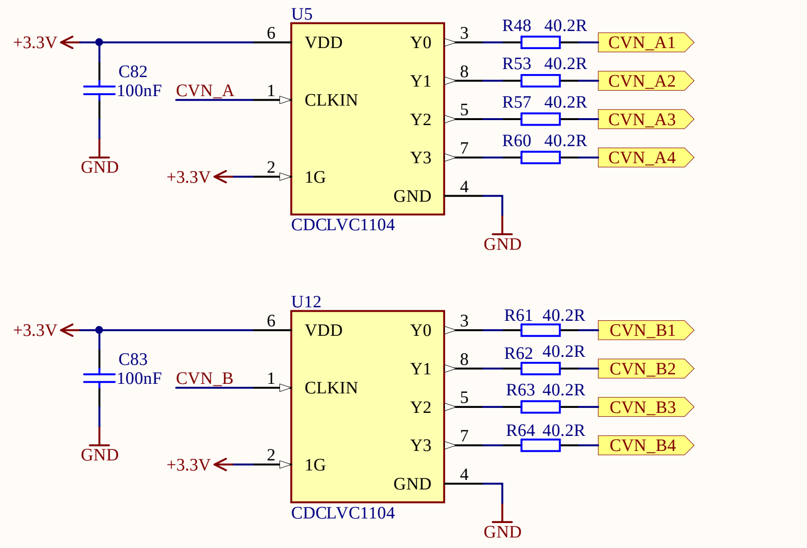
If R60 is fitted with 0R all ADCs are controlled with one conversion signal. R61 and R62 are then dnp

R50 and R51 (R61 and R62) are used to adjust 3.3V CVN signal to 2.5V level

R52, R53 and R54 are fitted apropriatly for Pilot In/Out

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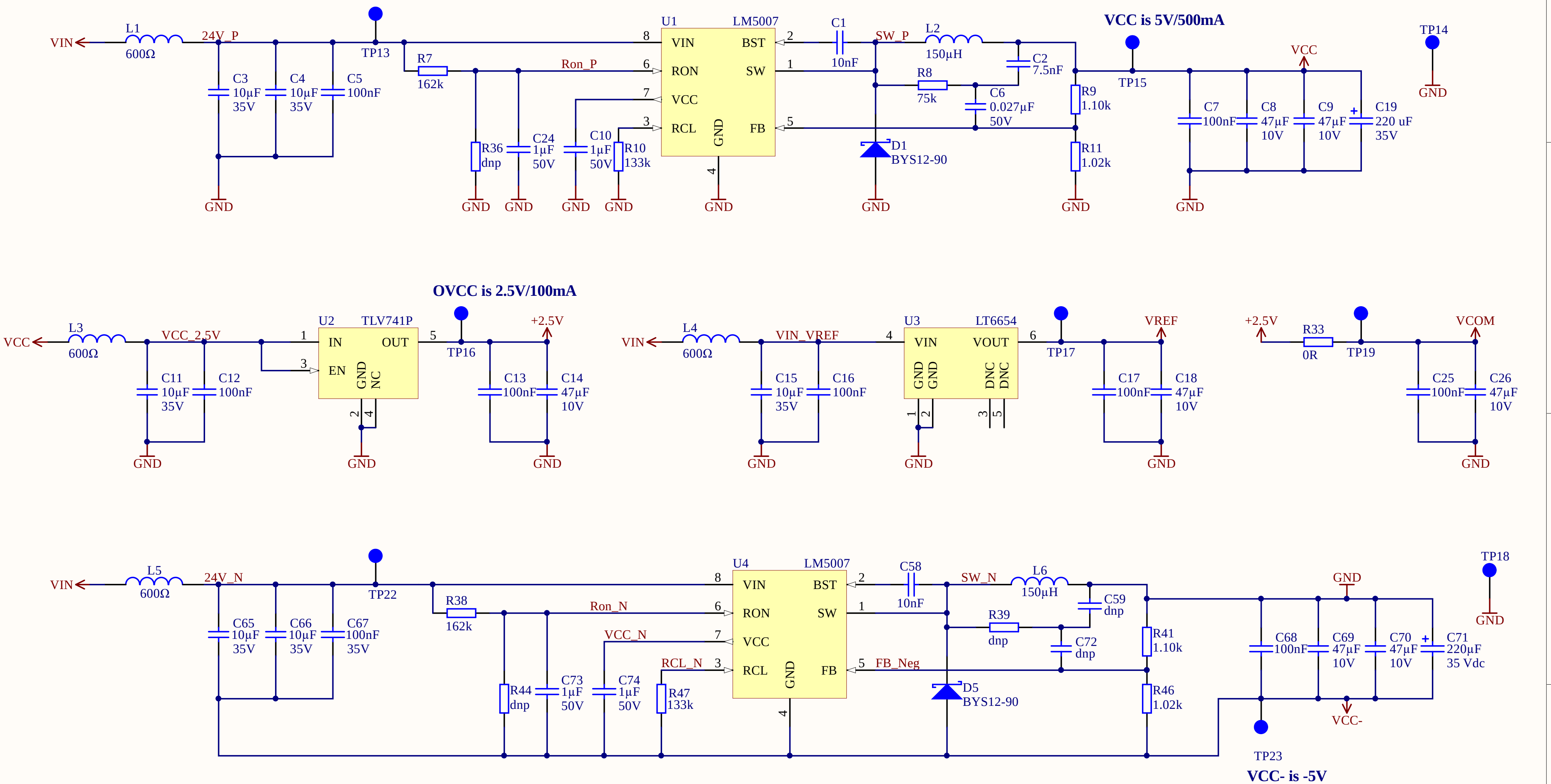
|---|---|---|
| Title: CarrierConnector.SchDoc | | TU München<br>Arcisstraße 21<br>80333 München<br>GERMANY |
| Size: A4 | Revision: 3v03 | |
| Date: 27.07.2020 | Time: 10:18:32 | Sheet 5 of 14 — Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | | |




LT6655 is available in 1.25V/2.048V/2.5V/3V/3.3V/4.096V/5V
If the internal reference from LTC2311 is used U3 must not be fitted

For +/-5.0V rails use 1kOhm resistors (CRCW06031K00JNEAC) for R6, R8, R12, R14

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| Size: A4 | Revision: 3v03 | | |
| Date: 27.07.2020 | Time: 10:18:32 | Sheet 6 of 14 | Author: Simon Lukas |
| Project: UltraZohm_ADC.PrjPCB | | | |